

Solutions for Week 11 Maths Exercise Sheet

Matrix Multiplication, Identity and Inverse

(1) Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

- (a) Find a 2×2 matrix B (if it exists) such that $BA = I$?
 (b) Find a 2×2 matrix C (if it exists) such that $AC = I$?

Comment: After answering part (a), is there an easy way to answer part (b)?

Solution:

- (a) Suppose there is a 2×2 matrix $B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ such that $BA = I$. We now compute BA using the matrix multiplication formula gives

$$BA = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} a+3b & 2a+4b \\ c+3d & 2c+4d \end{pmatrix}$$

Since $BA = I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ this gives us the following four equations (when two matrices are equal then each of their corresponding entries must be the same!)

$$a + 3b = 1 \tag{1}$$

$$2a + 4b = 0 \tag{2}$$

$$c + 3d = 0 \tag{3}$$

$$2c + 4d = 1 \tag{4}$$

We will first solve equation (1) and equation (2) to obtain values for a and b . Two times equation (1) gives

$$2a + 6b = 2 \tag{5}$$

Equation (5) minus equation (2) gives $2b = 2$ which implies $b = 1$. Substituting this value of b in equation (1) gives $a = -2$.

We now solve equation (3) and equation (4) to obtain values for c and d . Two times equation (3) gives

$$2c + 6d = 0 \tag{6}$$

Equation (6) minus equation (4) gives $2d = -1$ which implies $d = -\frac{1}{2}$. Substituting this value of d in equation (3) gives $c = \frac{3}{2}$.

Therefore, the matrix B which satisfies $BA = I$ is given by $\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{pmatrix}$

- (b) To solve this question, we could again do a similar approach to the previous part. But because of the following theorem (proof omitted!) which we mentioned in the lectures:

“If a matrix has a left inverse, then the right inverse also exists and is the same as the left inverse”

it follows that C also exists and is equal to the matrix B which we found in part (a).

(2) Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$ and $I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

- (a) Find a 3×3 matrix B (if it exists) such that $BA = I$?
 (b) Find a 3×3 matrix C (if it exists) such that $AC = I$?

Solution:

- (a) Suppose there is a 3×3 matrix $B = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & k \end{pmatrix}$ such that $BA = I$. We now compute BA using the matrix multiplication formula gives

$$BA = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & k \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} = \begin{pmatrix} a+4b+7c & 2a+5b+8c & 3a+6b+9c \\ d+4e+7f & 2d+5e+8f & 3d+6e+9f \\ g+4h+7k & 2g+5h+8k & 3g+6h+9k \end{pmatrix}$$

Since $BA = I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ this gives us nine equations (when two matrices are equal then each of their corresponding entries must be the same!). We only use the following three equations (corresponding to first row) to show that the matrix B cannot exist:

$$a + 4b + 7c = 1 \quad (7)$$

$$2a + 5b + 8c = 0 \quad (8)$$

$$3a + 6b + 9c = 0 \quad (9)$$

Equation (9) implies

$$a + 2b + 3c = 0 \quad (10)$$

Equation (7) minus equation (10) gives

$$5b + 4c = 1 \quad (11)$$

Equation (7) plus equation (10) minus equation (8) gives

$$b + 2c = 1 \quad (12)$$

Solving equation (11) and equation (12) gives $b = \frac{-1}{3}$ and $c = \frac{2}{3}$.

Substituting these values of b and c in equation (7) implies $a = \frac{-7}{3}$. But substituting these values of b and c in equation (8) implies $a = \frac{-11}{6}$. Contradiction. Hence, there are no values for a, b, c which satisfying each of the three equations given by equation (7), equation (8) and equation (9). Hence, the matrix B satisfying $BA = I$, i.e., the left-inverse of A , cannot exist.

- (b) This part asks if there is a (3×3) matrix C such that $AC = I$, i.e, does A have a right-inverse matrix C ? We claim that since we showed in part (a) that the left-inverse for A does not exist, then it follows from the following theorem (proof omitted!) which we mentioned in the lectures:

“If a matrix has a right inverse, then the left inverse also exists and is the same as the right inverse”

that the right-inverse matrix C for A also does not exist.

(3) Let $X = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 4 \\ 1 & 0 & 5 \end{pmatrix}$ and $I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

- (a) Find a 3×3 matrix Y (if it exists) such that $XY = I$?
 (b) Find a 3×3 matrix Z (if it exists) such that $ZX = I$?

Solution:

- (a) Suppose there is a 3×3 matrix $Y = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & k \end{pmatrix}$ such that $XY = I$. We now compute XY using the matrix multiplication formula gives

$$XY = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 4 \\ 1 & 0 & 5 \end{pmatrix} \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & k \end{pmatrix} = \begin{pmatrix} a+2d+3g & b+2e+3h & c+2f+3k \\ a+2d+4g & b+2e+4h & c+2f+4k \\ a+5g & b+5h & c+5k \end{pmatrix}$$

Since $XY = I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ this gives us nine equations (when two matrices are equal then each of their corresponding entries must be the same!) which are three sets of three linear equations in three unknowns. We solve each of them to find the matrix Y as follows:

- From the first column of $XY = I$ we get the following three equations in the variables a, d, g

$$a + 2d + 3g = 1 \quad (13)$$

$$a + 2d + 4g = 0 \quad (14)$$

$$a + 5g = 0 \quad (15)$$

Solving these three equations by Gaussian elimination gives $a = 5, d = \frac{-1}{2}$ and $g = -1$

- From the second column of $XY = I$ we get the following three equations in the variables b, e, h

$$b + 2e + 3h = 0 \quad (16)$$

$$b + 2e + 4h = 1 \quad (17)$$

$$b + 5h = 0 \quad (18)$$

Solving these three equations by Gaussian elimination gives $b = -5, e = 1$ and $h = 1$

- From the third column of $XY = I$ we get the following three equations in the variables c, f, k

$$c + 2f + 3k = 0 \quad (19)$$

$$c + 2f + 4k = 0 \quad (20)$$

$$c + 5k = 1 \quad (21)$$

Solving these three equations by Gaussian elimination gives $c = 1, f = \frac{-1}{2}$ and $k = 0$

Hence, we obtain the matrix $Y = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & k \end{pmatrix} = Y = \begin{pmatrix} 5 & -5 & 1 \\ \frac{-1}{2} & 1 & \frac{-1}{2} \\ -1 & 1 & 0 \end{pmatrix}$ which satisfies $XY = I$.

- (b) In part (a) of this question, we have found the right inverse Y of the matrix X . Hence, by the following theorem (proof omitted!) which we mentioned in the lectures:
“If a matrix has a right inverse, then the left inverse also exists and is the same as the right inverse”

it follows the left-inverse of X also exists and is the same as the matrix Y . Therefore, from part (a), we can just answer that $Z = Y = \begin{pmatrix} 5 & -5 & 1 \\ \frac{-1}{2} & 1 & \frac{-1}{2} \\ -1 & 1 & 0 \end{pmatrix}$ satisfies $ZX = I$.

- (4) Let $a, b, c, d \in \mathbb{Q}$ be such that none of them are zero. Let $X = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. If $ad - bc = 0$, then show that there is no 2×2 matrix Y whose entries are rational numbers and which satisfies $XY = I$.

Solution: Suppose that there is a 2×2 matrix $Y = \begin{pmatrix} r & s \\ t & u \end{pmatrix}$ with $r, s, t, u \in \mathbb{Q}$ and which satisfies $XY = I$. Then we have

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I = XY = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} r & s \\ t & u \end{pmatrix} = \begin{pmatrix} ar + bt & as + bu \\ cr + dt & cs + du \end{pmatrix}$$

This gives us the following four equations (when two matrices are equal then each of their corresponding entries must be the same!)

$$ar + bt = 1 \tag{22}$$

$$as + bu = 0 \tag{23}$$

$$cr + dt = 0 \tag{24}$$

$$cs + du = 1 \tag{25}$$

Multiplying equation (22) by d gives

$$dar + dbt = d \tag{26}$$

Multiplying equation (24) by b gives

$$bcr + bdt = 0 \tag{27}$$

Equation (26) minus equation (27) implies $d = r(ad - bc) = r \times 0 = 0$ since $ad - bc = 0$. This contradicts the given fact that none of the four rational numbers a, b, c, d are zero. Hence, the claimed matrix Y satisfying $XY = I$ cannot exist.